



Comparative Analysis of Native vs. Cross-Platform Mobile Development

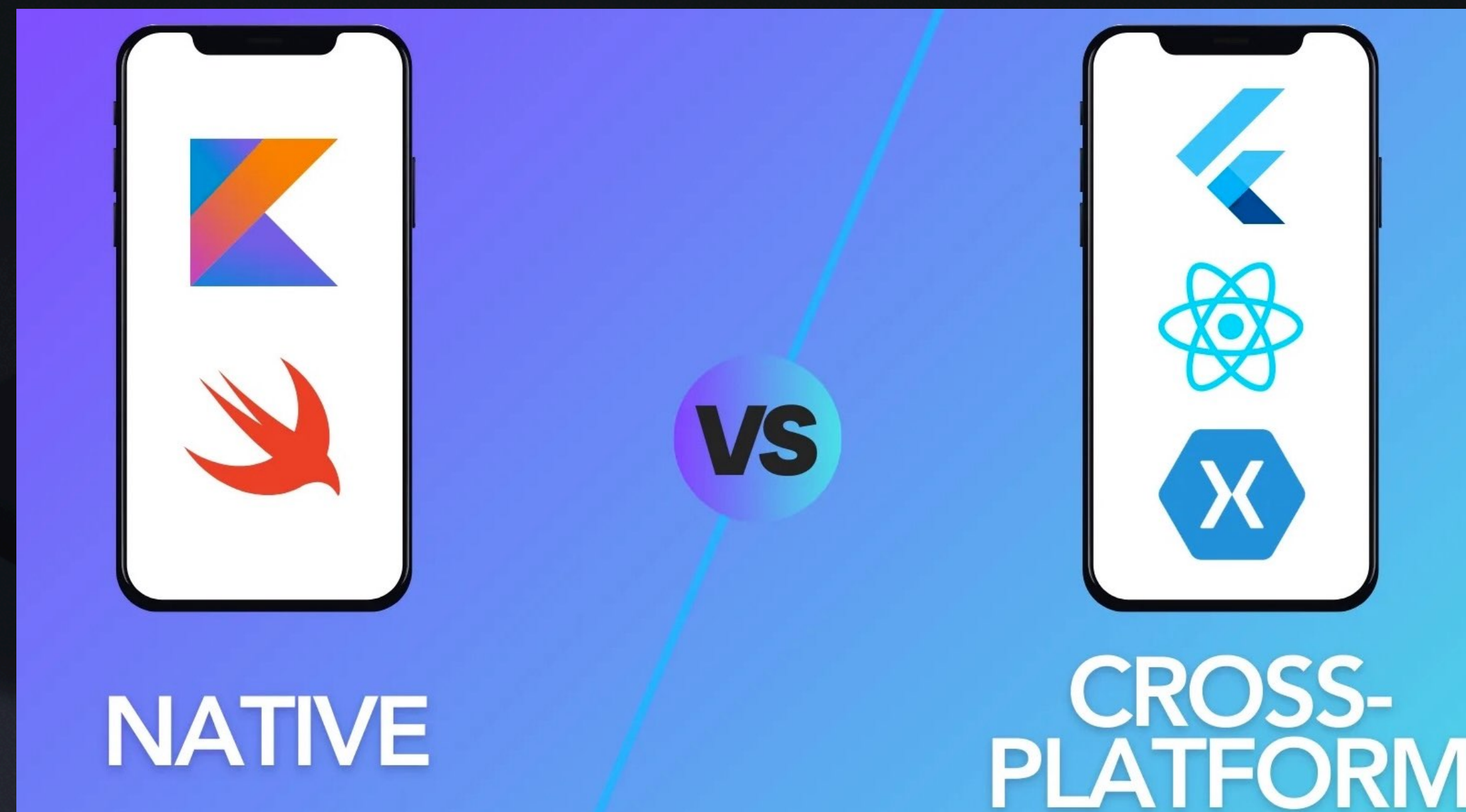
An Empirical Performance Study on iOS

Agenda

1. The Developer's Dilemma
2. The Research Gap
3. Methodology & Metrics
4. Experimental Results
5. The Trade-off
6. Conclusion

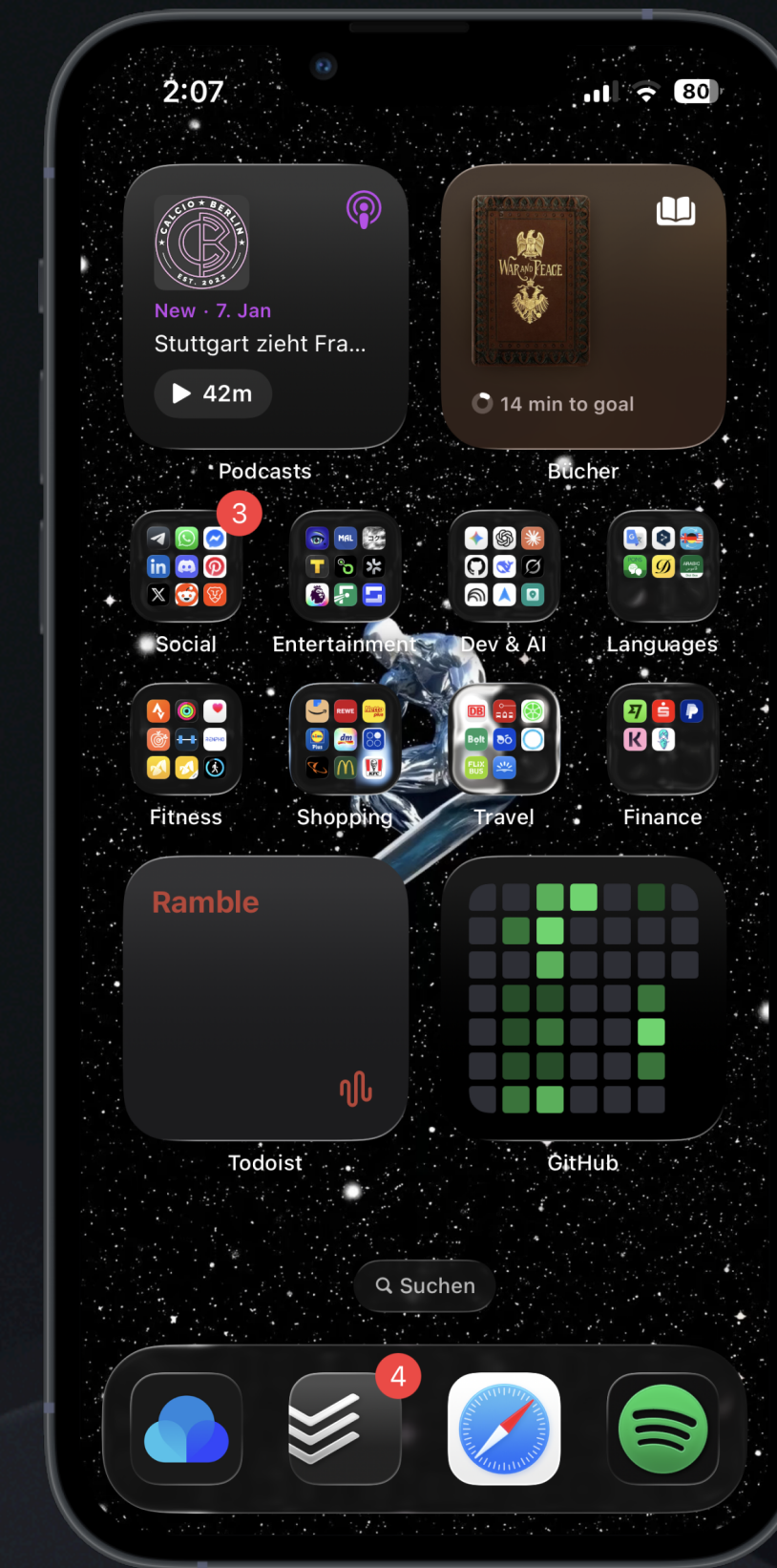
The Developer's Dilemma

- The Industry Challenge:
 - Native: High Performance, but High Cost (2 Codebases).
 - Cross-Platform (Flutter): "Write Once, Run Anywhere," but potential Performance Overhead.



The Research Gap

- Why this study?
- Existing data: Outdated or relies on Simulators/Android.
- My Focus: Physical Hardware (iPhone 14).
- Research Question: To what extent does the performance overhead of state-of-the-art cross-platform frameworks (Flutter) impact system resources compared to native development (Swift) on iOS devices?



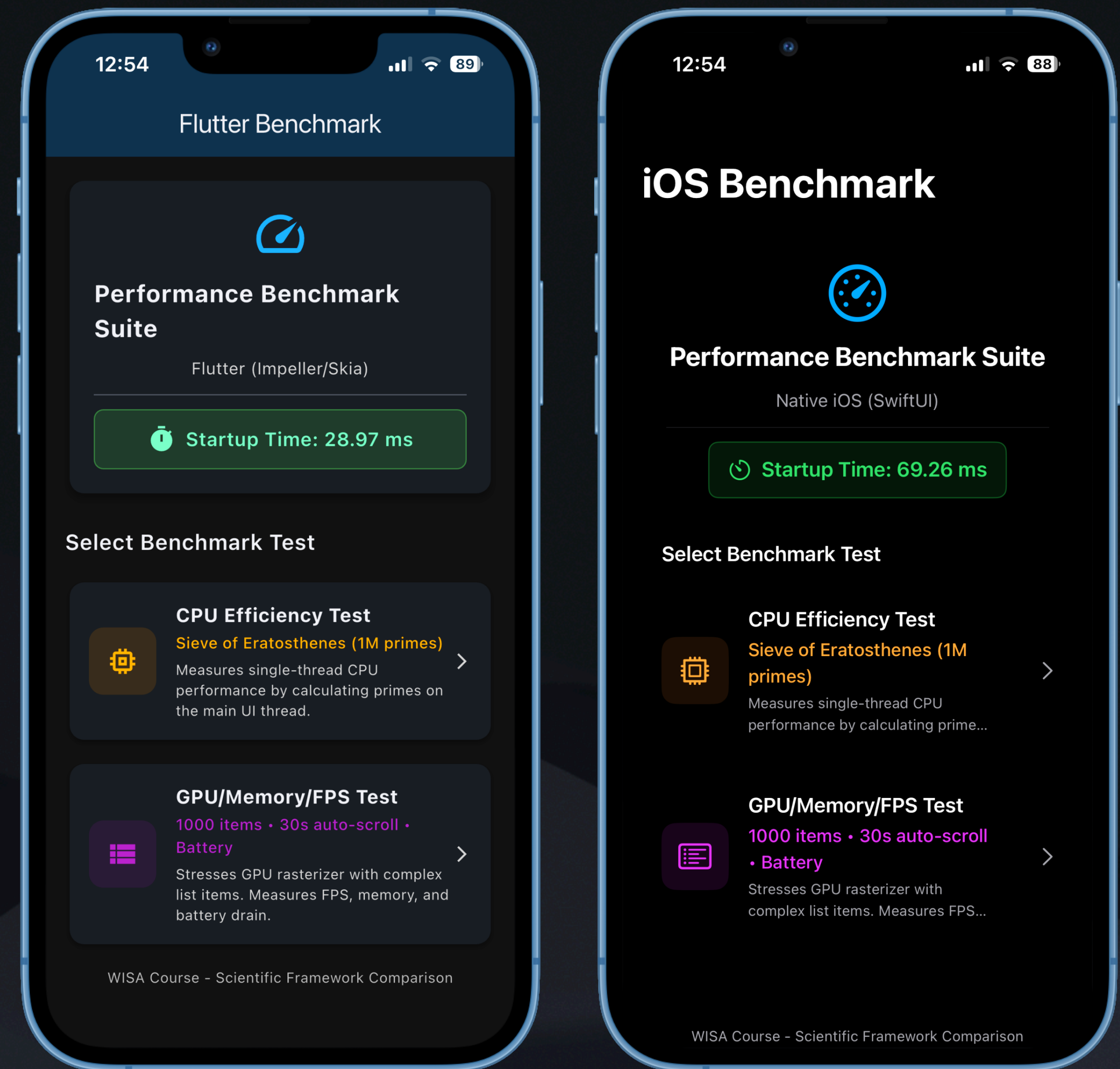
iPhone ✓



Simulator ✗

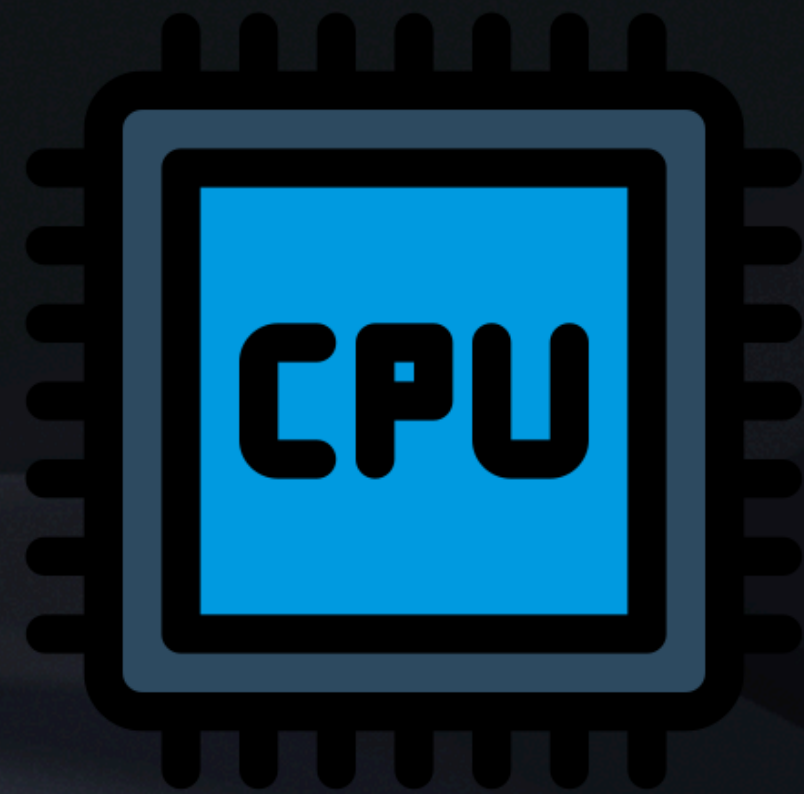
Methodology

- The Artifact: Two identical "Benchmark Apps".
- Native: Swift 6.2.3 (SwiftUI).
- Cross-Platform: Flutter 3.38.4 (Dart).
- Device: iPhone 14 (iOS 26)
- github.com/Milad9A/WISA-Cross-Platform-vs-iOS-Benchmark



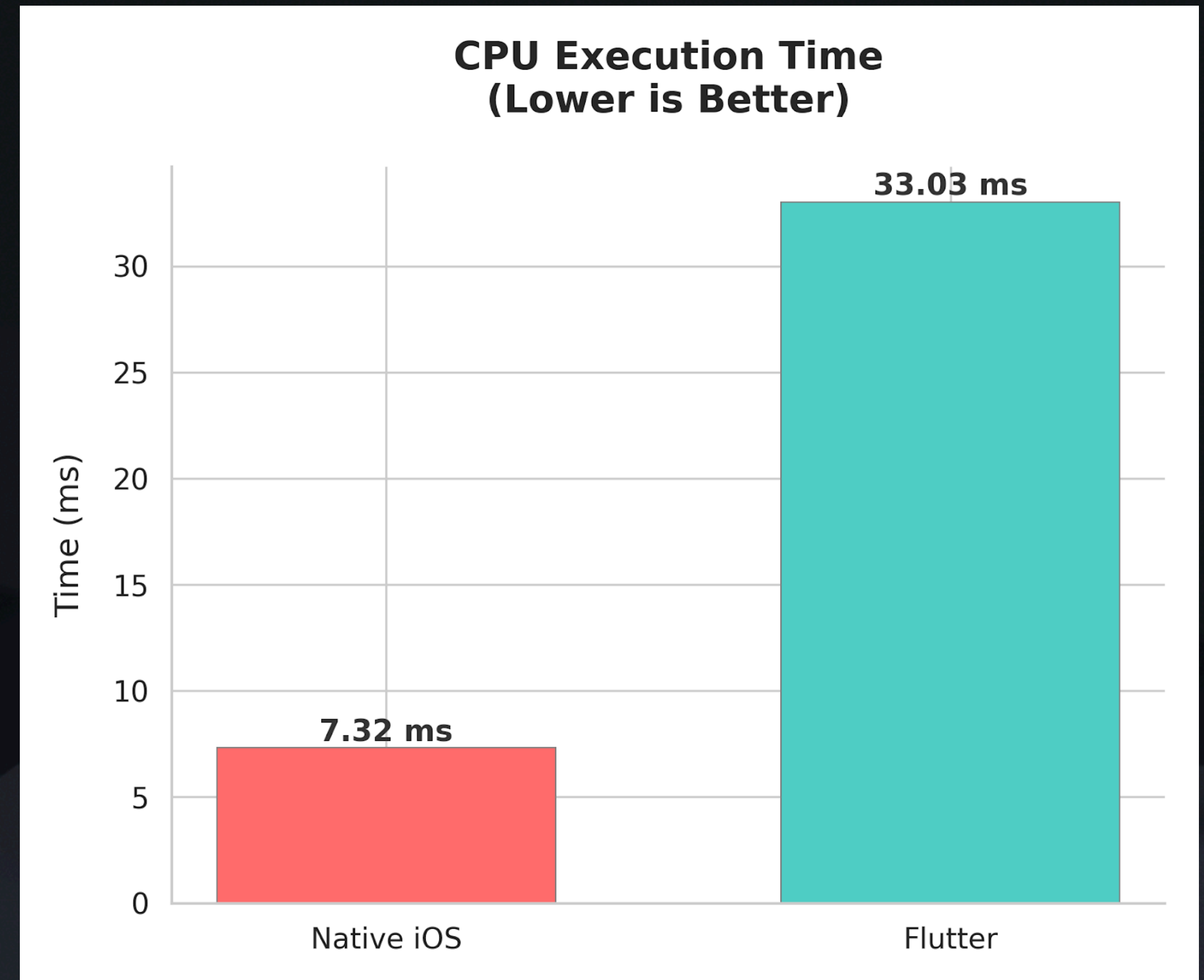
The Three Metrics

1. CPU Efficiency: Heavy calculation (Prime Numbers).
2. Rendering Stability: Scrolling smoothness (FPS & Jank).
3. Engine Initialization: "Framework-to-Frame" time.



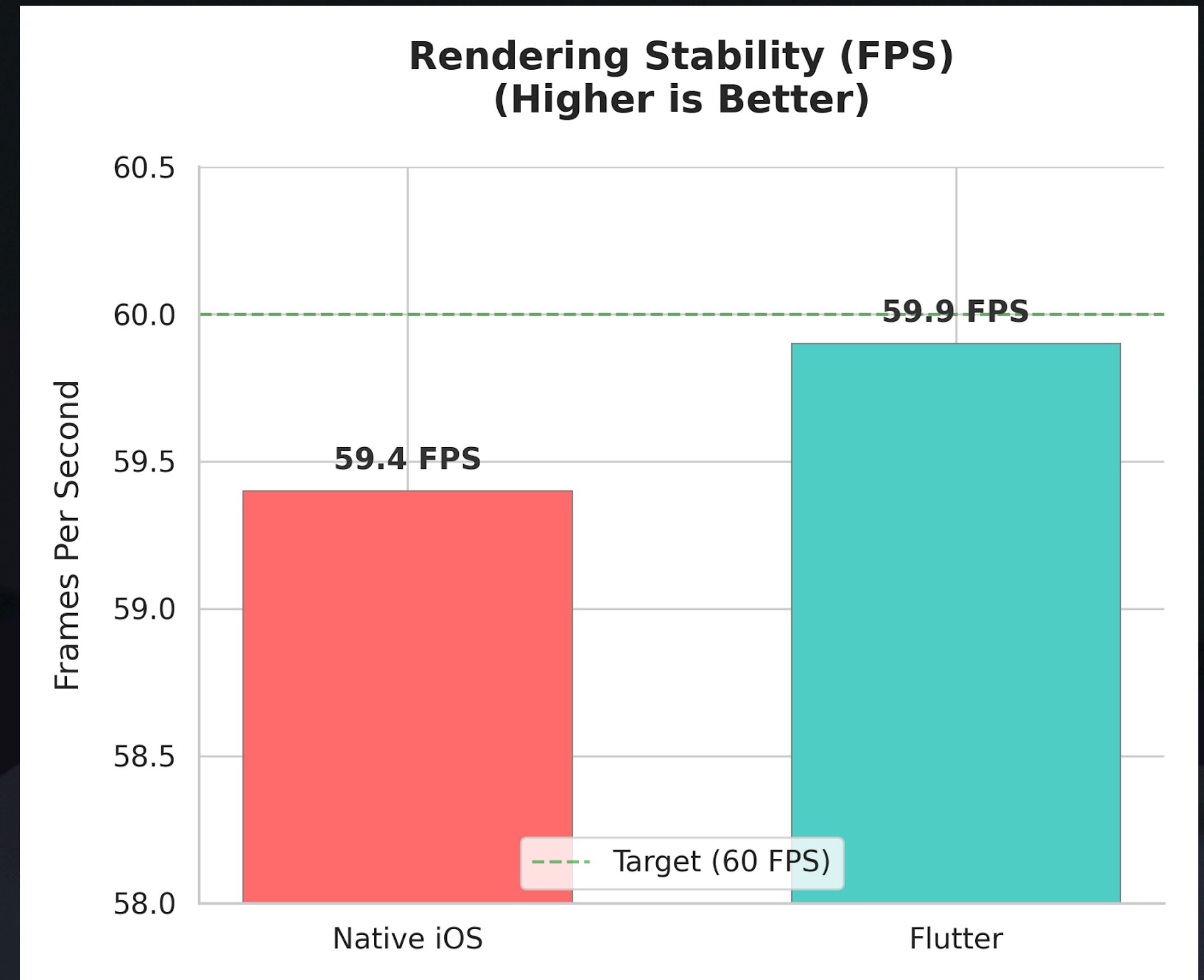
Result 1 - CPU Efficiency

- Task: Sieve of Eratosthenes (n=1M) on Main Thread.
- Native iOS: ~7.32 ms.
- Flutter: ~33.03 ms.
- Result: Native is ~4.5x faster.



Result 2 - Rendering Stability

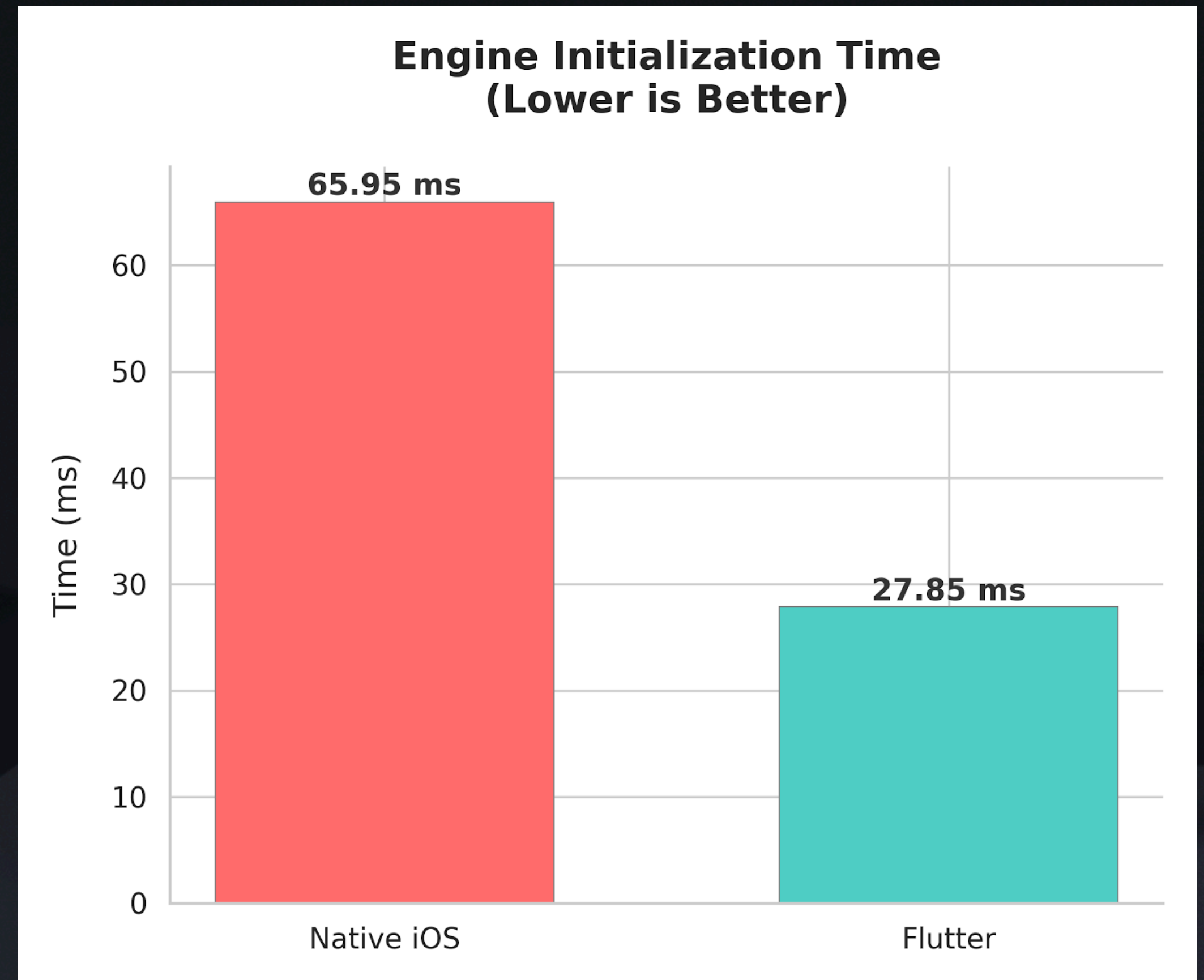
- Task: Scrolling 1,000 complex items.
- Target: 60 FPS.
- Flutter (Impeller): 59.9 FPS (0% Jank*).
- Native (SwiftUI): 59.4 FPS (0.5% Jank*).
- Impeller > Skia.



*Jank: Stuttering movement caused by uneven frame timing.

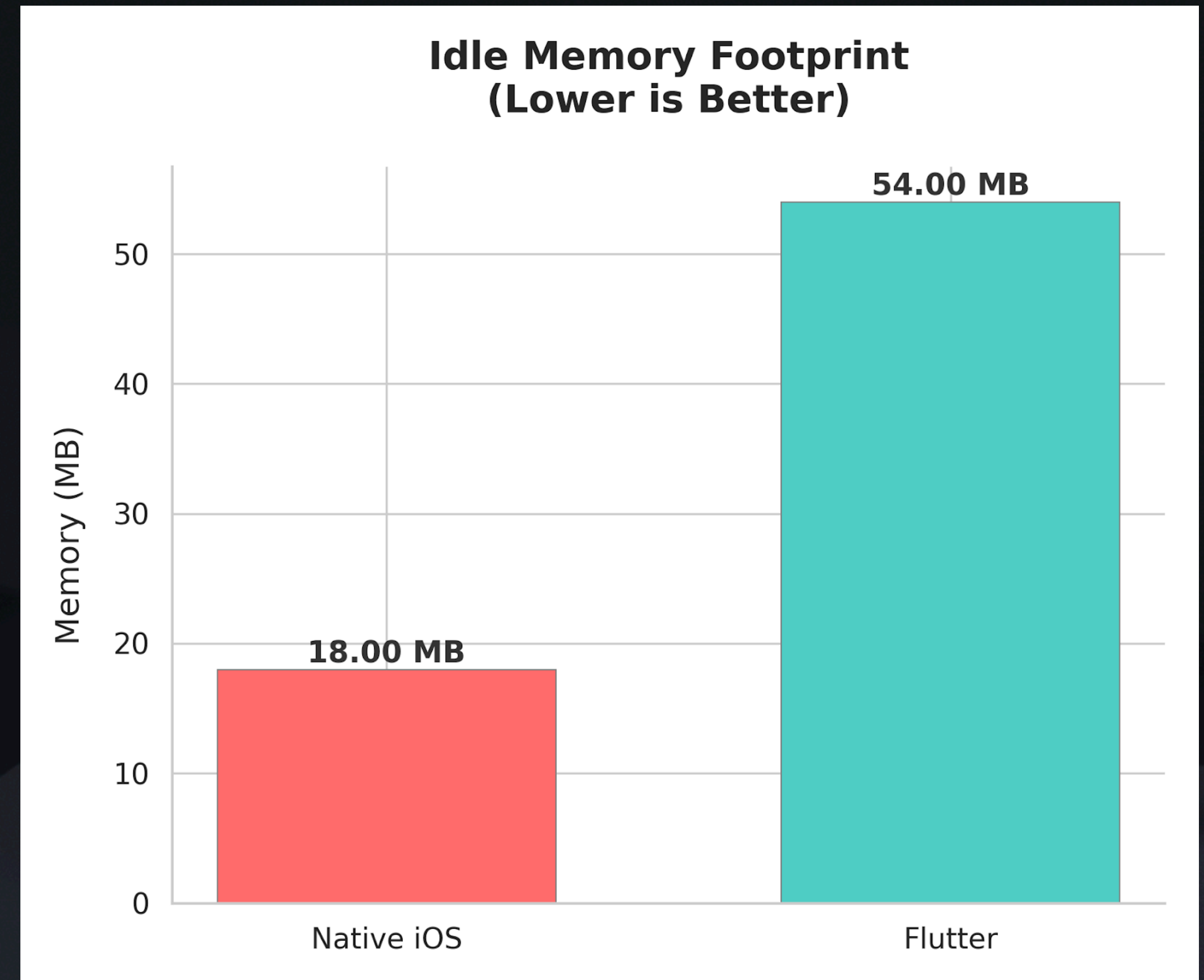
Result 3 - Initialization Time

- Metric: Time to first rasterized frame.
- Native iOS: ~66 ms.
- Flutter: ~28 ms.
- Result: Flutter launches ~2.4x faster.



The Trade-off: Memory & Size

- Idle Memory (RAM):
 - Native: 18 MB
 - Flutter: 54 MB (~3x higher)
- App Size:
 - Native: 2.8 MB
 - Flutter: 14.3 MB



Conclusion & Outlook

- Performance Gap: Narrowed significantly.
- Recommendation for 2026:
 - Use Native for: Heavy math, Cryptography, lightweight App Clips*.
 - Use Flutter for: Standard B2B/B2C apps.

Thank You!
Questions?